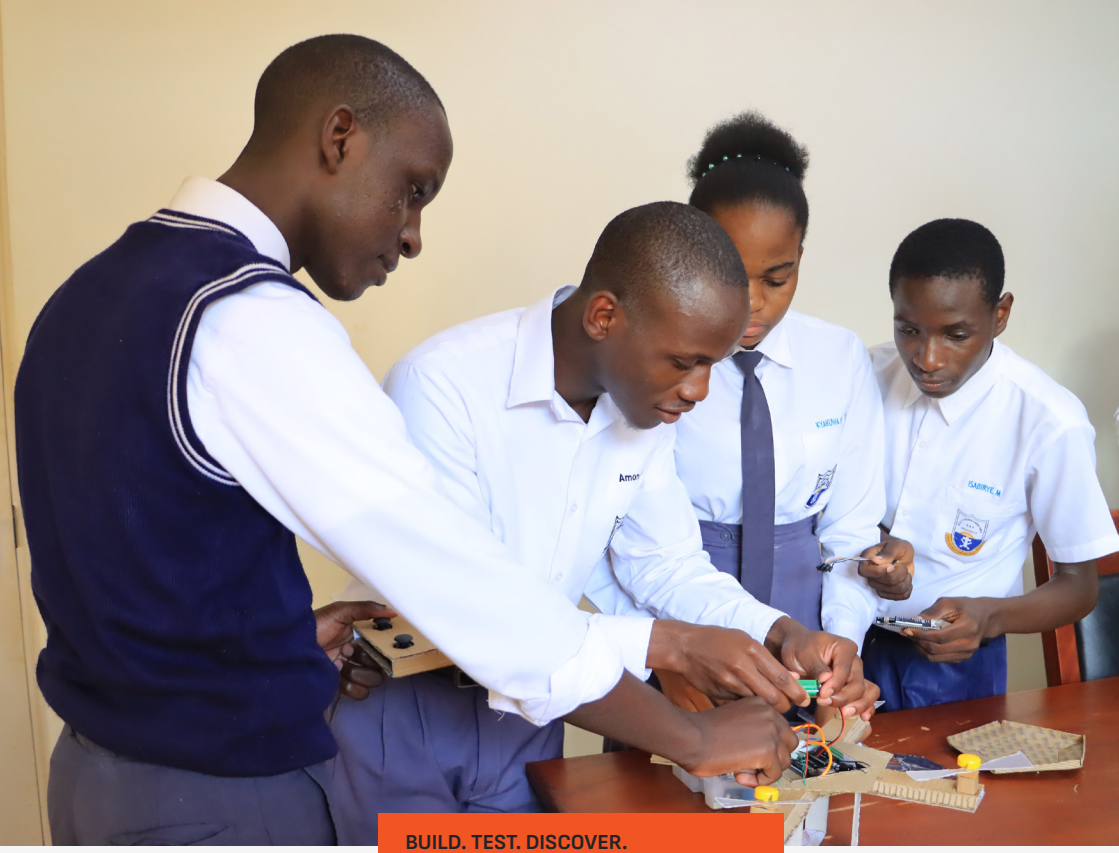


BUILD. TEST. DISCOVER.

JDIOBESTEM FOUNDATION

2026/27

STUDENT EDITION

STUDENT PROJECT

HANDBOOK

Student Project Proposal Workbook

Issued on: 03 September 2026

Prepared by: Joshua Mutesasira
Kampala, Uganda

Your project journey

01	Welcome and Purpose	4
02	About the Science Fair	6
03	Eligibility and Participation	8
04	Project Categories and Project Types	11
05	Choosing a Strong Project Idea	13
06	Understanding Research and Engineering Design	15
07	Writing the Project Proposal	17
08	Planning Objectives, Questions and Hypotheses	19
09	Methodology and Experimental Design	21
10	Data Collection, Analysis and Interpretation	23
11	Budgeting, Materials and Project Funding	25
12	Ethics, Plagiarism, and AI Use	27
13	Safety, Safeguarding & Risk Management	30

From evidence to impact

14	Human Participants and the Environment	33
15	Intellectual Property and Disclosure	35
16	Project Management and Documentation	37
17	Preparing the Final Report	39
18	Science Fair Display and Demonstration	41
19	Oral Presentation and Judge Interview	43
20	Judging Criteria and Scoring	45
21	Awards and Recognition	47
22	If Your Project Receives Foundation Funding	49
23	Common Reasons Projects Lose Marks	51
24	Frequently Asked Questions	53
25	Glossary	56

01 / Welcome and Purpose

1.1 Welcome to student innovators

The Jdiobe STEM Foundation Secondary School Science Fair helps students investigate questions, solve practical problems, build and test ideas, and explain evidence. The aim is to develop curiosity, skill and confidence through testing and revision.

Start with a problem at school or in your community, a question, a pattern in data, or an idea for improving a tool. Judges look for thoughtful decisions, evidence, learning and an explanation in your own words.

1.2 Purpose of the handbook

Use this handbook to check eligibility, choose a project, write a proposal, work safely and ethically, document results, and prepare for judging.

1.3 What this competition values

- Curiosity and willingness to ask meaningful questions.
- Original thinking and thoughtful improvement of existing ideas.
- Use of science, technology, engineering and mathematics to understand or solve problems.
- Evidence based reasoning rather than unsupported claims.
- Student ownership of the work.
- Safe and ethical conduct.
- Practical relevance to schools, communities, Uganda, Africa or the wider world.
- Clear communication, including honest discussion of limitations and failed attempts.
- Resourcefulness. Excellent projects do not have to be expensive.

- Learning. A project that teaches you something important can be valuable even when the original idea does not work exactly as expected.

Remember

amount of money spent. A carefully designed low cost project can be stronger than an expensive project with weak reasoning.

PUT IT INTO PRACTICE

A question I want to investigate

_____

_____

Why it matters to my school or community

_____

_____

02 / About the Science Fair

2.1 Overall aims

The fair connects classroom learning with real problems and gives students a place to present their work to educators, STEM professionals and the community.

2.2 Learning outcomes

A project builds these skills:

- Defining a problem clearly.
- Formulating a research question or engineering challenge.
- Searching for reliable background information.
- Setting realistic objectives.
- Planning fair tests and controlled comparisons.
- Collecting data systematically.
- Making conclusions that match the evidence.
- Preparing budgets and timelines.
- Identifying risks and working safely.
- Using sources responsibly.
- Presenting technical work to different audiences.
- Responding to questions and constructive criticism.

Stage	What happens
Idea	Student identifies a question, need, challenge or opportunity.
School discussion	Student discusses feasibility, safety and supervision with a teacher mentor.

Stage	What happens
Proposal	Student prepares and submits the official project proposal.
Review	Proposal is checked for eligibility, quality, safety, ethics and feasibility.
Development	Approved students perform research, experiments, design, coding, construction or analysis.
Documentation	Students keep a project notebook, records, photographs where allowed, data and receipts where funding applies.
School level preparation	Students refine their report, poster, prototype and oral explanation.
Science fair	Selected projects are displayed and evaluated by judges.
Recognition and next steps	Awards, feedback, mentorship or additional support may be offered according to the current competition cycle.

Plan Backwards

start late run out of time for pilot testing and revision — the two stages judges notice most.

03 / Eligibility and Participation

3.1 Basic eligibility

A student may apply when all of the following conditions are met:

- The student is enrolled at a school or institution participating with Jdiobe STEM Foundation.
- The student is within the eligible year groups or levels stated in the current competition announcement: All levels.
- The project uses one or more STEM disciplines or meaningfully contributes to STEM learning.
- The project can be completed within the official project period: six months.
- A teacher at the institution agrees to mentor and supervise the project.
- The institution authorizes the application.
- The student agrees to follow all safety, ethics, integrity and science fair rules.

3.2 Individual and team projects

Students may apply individually or in teams of up to five students. Each team must name one student as the primary contact. All team members must make a meaningful intellectual and practical contribution. A team should not include a student only so that the student can appear on the final project.

Team responsibility

- Every team member must understand the entire project, not only one assigned part.
- The team should agree how decisions will be made and how work will be shared.
- The project notebook should record major contributions by each student.

- During judging, any team member may be asked about any part of the project.
- If a team changes membership, the teacher mentor should document the change and notify the Foundation when required.

3.3 Teacher mentor

Every project must have a teacher mentor approved by the school. The mentor helps students understand scientific and engineering practice, asks questions, checks safety, helps students access school facilities and confirms that the work is genuinely student led. The mentor must not write the proposal, fabricate data, perform the project for the student or prepare answers for judges.

3.4 School authorization

Participation occurs through the school or institution. The school must authorize the application regardless of the student's age. The school is also responsible for day to day supervision of project activities carried out under its authority.

3.5 Students aged 18 or above

Students aged 18 or above participate through their institution in the same way as other students. Additional safeguarding or institutional requirements may apply when an adult student works with participants under 18. The institution must determine and arrange any required checks before such work begins.

Five students, not five names

What a mentor must not do

recording data or rehearsing answers for judges crosses the line, and reviewers can usually tell.

04 / Project Categories and Project Types

4.1 Choosing the best category

Select the category that best represents the main question, method or solution in your project. A project may use several disciplines, but it should have one primary category for judging. The Foundation may move a project to a different category when that would result in a fairer technical review.

Category	Examples
Engineering and Design	Structures, machines, transport, civil systems, manufacturing, materials, product design, water systems and practical engineered solutions.
Robotics and Automation	Robots, control systems, sensors, autonomous systems, mechatronics and automated processes.
Computer Science and Artificial Intelligence	Software, algorithms, applications, machine learning, computer vision, cybersecurity concepts, data systems and responsible AI applications.
Physical Sciences	Physics, chemistry, astronomy, materials, energy, electricity, mechanics, waves, heat and related investigations.
Biological and Life Sciences	Plant science, microbiology at approved safe levels, ecology, genetics concepts, physiology and other life science investigations.
Environmental Science and Climate	Water, air, soil, waste, conservation, climate resilience, ecosystems, pollution and environmental monitoring.
Agriculture and Food Security	Crop production, irrigation, soil, post harvest systems, food preservation, livestock related non invasive studies and agricultural technology.

Category	Examples
Health and Biomedical Innovation	Public health, non diagnostic health technology, assistive concepts, biomedical engineering and health education studies that meet ethics requirements.
Mathematics and Data Science	Mathematical models, optimization, probability, statistics, data analysis and mathematical investigations.
Renewable Energy and Sustainability	Solar, wind, biomass, energy storage, efficiency, clean cooking, conservation and sustainable systems.
STEM Education and Community Innovation	Projects that test or create improved ways to teach STEM or apply STEM knowledge to a clearly defined community need.

05 / Choosing a Strong Project Idea

5.1 Start with observation

Observe your school, home and community. What wastes resources? What is difficult to measure? What could be safer, cheaper, cleaner or more reliable? Begin with a question you can investigate.

- Specific enough to investigate within the available time.
- Connected to a meaningful question or need.
- Suitable for the student's current knowledge with room to learn.
- Possible using available or realistically obtainable materials.
- Safe and ethically acceptable.
- Capable of producing evidence that can be measured, observed or evaluated.
- Original in the sense that the student makes real decisions and contributes something beyond simply repeating instructions.
- Able to be explained clearly to another student, teacher and judge.

5.3 Originality does not mean inventing something nobody has ever seen

Original work can improve an existing design, apply a known method to a local problem, compare approaches, reduce cost or collect new data. Acknowledge earlier work and explain your contribution.

5.2 Characteristics of a strong idea

5.4 Ideas to avoid

- Demonstrations that only show a known effect without asking a question or collecting meaningful data.
- Ready made kits assembled exactly according to instructions with no modification, testing or analysis.
- Projects copied from websites, videos, previous competitions or another student.

- Projects whose only claim is that a product works, with no defined test.
- Projects that are so broad that the student cannot complete them.
- Projects requiring dangerous materials, unsafe procedures or unapproved human testing.
- Projects that promise unrealistic impact without evidence.

Avoid category shopping

need the correct expertise to evaluate you fairly.

06 / Understanding Research and Engineering Design

6.1 The scientific method as a flexible process

An investigation needs a clear question, background research, a suitable method, careful measurements, analysis and reflection. Include a reasoned prediction when appropriate. If evidence changes your approach, record what changed and why.

6.2 Engineering design process

1. Define the need or problem.
2. Identify users or beneficiaries.
3. Write measurable requirements and important constraints.
4. Research existing approaches.
5. Generate several possible concepts.
6. Compare concepts using reasonable criteria.
7. Select and justify a design.
8. Build a model, prototype or simulation.
9. Test against the requirements.
10. Analyze failures and weaknesses.
11. Improve the design.
12. Retest and document what changed.

6.3 Variables

Term	Meaning	Example
Independent variable	The factor deliberately changed by the researcher.	Amount of natural coagulant added to equal water samples.
Dependent variable	The outcome measured.	Turbidity after treatment.
Controlled variables	Factors kept as similar as practical so the comparison is fair.	Water volume, settling time, container size and measurement method.
Confounding factor	A factor that may influence the result and make interpretation harder.	Different starting water quality between samples.

6.4 Replication and repeated trials

Repeat tests where practical and report the number of trials. Repetition reveals variation and measurement error, but it cannot fix a poor method. First make the procedure fair and consistent.

Example

Instead of saying "build a low cost water level alarm," define what low cost means, the minimum detectable water level, the expected alarm distance, acceptable power source, maximum prototype budget and conditions where the device should operate.

07 / Writing the Project Proposal

7.1 Why the proposal matters

A proposal explains your plan before the main work begins. It lets reviewers assess clarity, feasibility, safety, ethics and measurable outcomes. Use precise, simple language.

7.2 Student authorship

Students must write their own proposal. A teacher may explain concepts, suggest questions, point out unclear sections and correct obvious formatting problems, but the teacher must not replace the student's thinking or write the application. If reviewers believe the proposal was written by someone else, they may request clarification or reject the application.

7.3 Recommended proposal sections

Section	What reviewers need to see
Project title	A short, specific title that tells the reader what the project investigates or designs.
Student and school information	Names, level, school, district, contact route through the school and teacher mentor.
Project category	The primary competition category.
Problem statement	The exact problem, need or unanswered question.
Background and rationale	What is already known and why this project is worth doing.
Research question or design challenge	The main question to answer or performance problem to solve.
Objectives	Specific results the student plans to achieve.

Section	What reviewers need to see
Hypothesis	A reasoned prediction for experiments where a hypothesis is appropriate.
Methodology	What will be done, with enough detail for reviewers to understand the process.
Materials and equipment	Main items needed.
Data and evaluation plan	What will be measured, how often, and how results will be analyzed.
Risk and ethics review	Potential hazards and how they will be controlled.
Timeline	Major activities and target completion dates.
Budget	Itemized costs and source of estimates.
Expected results and impact	What change or knowledge may result.
Sustainability or next steps	What may continue after the competition.
References	Sources used to understand the topic and plan the project.
Student declaration and approvals	Confirmation of originality, teacher review and institutional authorization.

Precision

Reviewers are not impressed by long words. Replace “improve water quality” with the measurement you will take, the conditions you will test and the size of change you expect to see.

08 / Planning Objectives, Questions and Hypotheses

8.1 Research question

A research question should be specific enough to answer using evidence. Good questions often identify what will be changed or compared, what outcome will be measured and the conditions of the study.

Examples

- How does the amount of shading affect the electrical output of a small solar panel under the same test conditions?
- Which of three locally available filtration media combinations produces the greatest reduction in measured turbidity for the selected water source?
- How accurately can a low cost sensor system detect classroom occupancy compared with manual counts?

8.2 Engineering challenge statement

Engineering projects may use a challenge statement rather than a traditional research question. The statement should define the user need and measurable performance targets.

8.3 Objectives

Objectives describe what the project will accomplish. Strong objectives are specific, measurable and realistic. Avoid listing activities such as “buy materials” as major objectives. Buying materials is a task. An objective describes a result.

Objective Test

- Does the objective contain an action that can be demonstrated or measured?
- Does it identify what will be studied, built, compared or evaluated?
- Can it reasonably be achieved during the project period?
- Will the data or final prototype show whether it was achieved?

8.4 Hypothesis

A hypothesis is a testable prediction based on prior knowledge. Not every project needs one. Descriptive studies, many engineering projects, software projects and exploratory mathematical investigations may not use a formal hypothesis. Do not invent a hypothesis simply because you think science projects must have one.

Engineering Challenge

Design and test a low cost hand washing water dispenser for a school setting that can deliver a controlled amount of water per use, operate without mains electricity and be constructed using materials available within the approved budget.

09 / Methodology and Experimental Design

9.1 Purpose of the methodology

The methodology tells another knowledgeable person exactly how you plan to carry out the project. It should be detailed enough to evaluate whether the process is fair, safe and capable of answering the research question or testing the design.

9.2 What to include

- Study or test location.
- Materials, equipment, software or datasets.
- Sample size or number of trials.
- Variables and controls.
- Step by step procedure.
- Measurement instruments and units.
- Calibration or consistency checks where appropriate.
- How results will be recorded.
- How data will be analyzed.
- How safety risks will be controlled.
- How waste or materials will be disposed of safely.
- What will happen if the first method does not work.

9.3 Fair testing

When comparing conditions, change only the factor you intend to study as far as practical. Keep other important conditions consistent and record unavoidable differences. A perfect laboratory may not be available, but you must show that you understand what could affect your results.

9.4 Controls and comparison groups

A control or baseline helps show whether the tested intervention actually makes a difference. The correct type of control depends on the project. A water treatment study may compare treated samples with an untreated

sample. A software project may compare a new method with a simple baseline algorithm. An engineering project may compare the improved prototype with the original design or required performance threshold.

9.5 Measurement quality

- Use appropriate units and record them consistently.
- Do not report more decimal places than the instrument can actually measure.
- When using a homemade measurement system, explain how you checked it.
- Record unexpected readings rather than deleting them automatically.
- If you remove a data point, explain the reason.
- Take photographs of setups when permitted, but photographs do not replace measurements.

9.6 Pilot testing

A small pilot test can reveal problems before the full experiment begins. Use it to check whether equipment works, instructions are clear, data sheets make sense and the planned measurement range is appropriate. Record what you changed after the pilot.

The stranger test



10 / Data Collection, Analysis and Interpretation

10.1 Data must be traceable

Record data when it is collected, not later from memory. Every data table should identify the date or trial, measurement, unit and relevant condition. Never change a result because it does not match your expectation.

10.2 Quantitative and qualitative data

Quantitative data are numerical, such as temperature, mass, time, voltage, count, percentage or test score. Qualitative data describe characteristics or observations, such as visible color change, failure mode or user feedback. A project may use both.

10.3 Basic analysis

- Organize raw data in tables.
- Check for missing values or obvious recording errors.
- Calculate suitable summaries such as totals, means, medians, ranges, rates or percentages where appropriate.
- Use graphs that match the type of data.
- Label axes and include units.
- Compare groups or conditions.
- Describe patterns and exceptions.
- Where advanced statistics are used, explain why the method is suitable rather than including a test you do not understand.

10.4 Correlation is not automatically causation

If two variables change together, that does not prove that one caused the other. Other factors may explain the relationship. Be careful with statements such as “X causes Y” unless your design provides strong evidence for a causal

claim.

10.5 Negative and unexpected results

A result that does not support your hypothesis is still a scientific result. Do not hide it. Explain what happened, what the evidence suggests, what limitations may have affected the study and what you would test next.

10.6 Conclusions

A conclusion should answer the research question or state how well the design met its requirements. It should refer to actual results. Avoid claiming that a small school experiment proves something for all of Uganda or for all people. State the scope of your evidence honestly.

Record it now

Claim only what you tested

"In the three classrooms tested" is stronger than "in Uganda." State the scope of your evidence honestly and a judge can verify it. Overclaim once

11 / Budgeting, Materials and Project Funding

11.1 General principle

Ask for what the project genuinely needs and no more. A smaller, clearly justified budget is stronger than a large vague request. Reviewers look for evidence that prices are realistic and that the project can continue if one item becomes unavailable.

11.2 Eligible costs

Subject to the official competition announcement, appropriate project expenses may include consumable materials, basic components, sensors, fabrication materials, approved laboratory supplies, printing directly needed for the project, data collection materials and other reasonable costs that are necessary to conduct the work.

11.3 Costs normally not supported

- School fees.
- A personal laptop or phone for the student.
- Payment or wages for student team members.
- Projects already completed before approval.
- Travel that is not genuinely required for the project.
- Religious or political activities.
- Equipment for which no responsible school custodian or future use plan is identified.
- Unexplained contingency amounts.
- Luxury versions of items when a reasonable lower cost option would meet the project need.

11.4 Building a budget

Budget element	Good practice
Item description	Name exactly what will be purchased.
Quantity	State how many units are needed.
Unit cost	Use a recent real price where possible.
Source	Name the shop, supplier or quotation source when available.
Total	Quantity multiplied by unit cost.
Justification	Explain why the item is necessary.
In kind contribution	Record items or services provided free by the school or another supporter.

Budget honesty

Do not round prices upward “just in case.” If a contingency is permitted in the current funding cycle, explain the percentage or amount and what risk it covers.

12 / Ethics, Plagiarism, and AI Use

12.1 Scientific integrity

Integrity means that the work presented is a truthful record of what you actually did, observed, calculated and concluded. Judges must be able to trust your project.

12.2 Prohibited conduct

- Inventing or changing data.
- Copying text, code, figures, designs or ideas without acknowledgment.
- Submitting another person's project as your own.
- Claiming tests were performed when they were not.
- Hiding important failed tests when they change the meaning of the results.
- Changing dates or receipts to obtain funding.
- Using photographs or results from another project as if they were your own.
- Allowing a teacher, parent, professional or AI system to produce work that you then present as your own understanding.

12.3 Plagiarism

Plagiarism occurs when you use another person's words, ideas, images, designs, code or other work without appropriate acknowledgment. Copying may occur even if only a few words are changed. Use your own words to explain what you learned and cite the source.

12.4 Collaboration and outside help

Students may receive reasonable help from teachers, laboratory staff, family members, universities, companies or other mentors, but that help must be disclosed when it materially affects the project. A mentor may teach you to use an instrument. A technician may cut a material that a student is not

permitted to cut. A programmer may explain an algorithm. The student must still make the project decisions and understand the resulting work.

12.5 Responsible use of artificial intelligence

Artificial intelligence tools may be useful for learning, brainstorming, language support, debugging and exploring possible approaches. They can also produce incorrect information, invented references, insecure code, biased outputs or writing that does not represent the student's understanding. For that reason, AI use must be transparent and limited to responsible assistance.

Acceptable examples

- Asking an AI tool to explain a scientific concept and then verifying the explanation using reliable sources.
- Using AI to suggest alternative search terms for background research.
- Using AI to help identify grammar problems in text the student wrote.
- Using AI to explain an error message in student written code.
- Using AI to brainstorm possible test variables, followed by Unacceptable examples
- Submitting an AI generated proposal or report as student written work.
- Asking AI to invent observations, measurements, survey results or references.
- Using AI generated code that the student cannot explain.
- Using AI to fabricate photographs, experiment images or evidence.
- Allowing AI to make the central scientific interpretation without student verification.

AI disclosure



13 / Safety, Safeguarding & Risk Management

13.1 School responsibility and supervision

Students participate through their school or institution. The institution is responsible for the welfare and supervision of students while project activities are carried out under its authority. A teacher must know where students are and what they are doing whenever they work on the project.

13.2 General safety principles

- Identify hazards before beginning work.
- Use the least hazardous method that can answer the question.
- Follow school laboratory and workshop rules.
- Use suitable personal protective equipment when required.
- Do not work alone on activities that require supervision.
- Do not bypass safety guards or electrical protection.
- Stop work when conditions become unsafe.
- Store tools, chemicals and equipment securely.
- Dispose of waste responsibly.
- Report injuries, spills, equipment failures and near misses to the teacher mentor immediately.

13.3 Risk assessment

Every project should consider hazards, who could be harmed, how serious the harm could be, how likely it is, and what controls will reduce the risk. Projects involving chemicals, heat, pressure, electricity, moving machinery, biological material, field work, vehicles, water bodies, heights or vulnerable people require particular attention.

13.4 Safeguarding rules

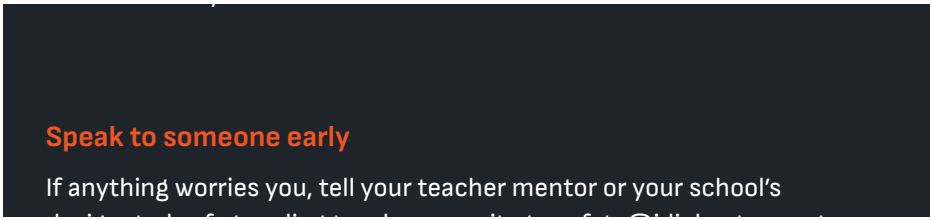
- Your institution must authorize the application before submission.

- If something worries you, speak to your teacher mentor or your institution's designated safeguarding teacher. Jdiobe STEM Foundation safeguarding contact: safety@jdiobestem.org.
- Never travel alone to meet someone about the project. Follow school approved travel and supervision arrangements.
- Never publish your home address or personal phone number as part of the project.
- Do not photograph or identify younger pupils without the institution's written approval and any required consent.
- An adult from the institution must be present when a project involves teaching or working with younger pupils. A student must not be left alone with younger children.

13.5 Projects that may require rejection or major modification

A project may be rejected, paused or modified when the risk cannot be managed appropriately in the school setting. Scientific interest does not override safety. The Foundation and school may require additional expert supervision or prohibit an activity entirely.

Three questions, every session





PUT IT INTO PRACTICE

The main risks in my project

Who must review my safety plan

14 / Human Participants and the Environment

14.1 Human participants

Projects involving surveys, interviews, health information, testing people, measuring physical performance, collecting personal data or asking sensitive questions require additional care. Students must not begin such activities until the school has determined what permissions and consent are required and the project has received any approval required by the competition.

Minimum principles

- Participation must be voluntary.
- Participants must understand what they are being asked to do.
- Students should collect only the information genuinely needed.
- Private information must be protected.
- Children and vulnerable participants require additional safeguards.
- Do not collect medical samples or perform invasive procedures.
- Do not claim to diagnose or treat a medical condition.
- Questions about trauma, abuse, illegal activity, sexual behavior or other highly sensitive matters should not be used in ordinary student projects without formal expert review.

14.2 Animals

Projects must treat animals humanely and comply with school and competition requirements. Procedures that intentionally cause pain, injury, distress, deprivation or death are not acceptable for ordinary student science fair work. Observation of normal behavior or non invasive environmental studies may be appropriate when properly supervised.

14.3 Microorganisms and biological materials

Culturing unknown environmental microorganisms can create hazards because the student may grow harmful organisms without knowing it. Such

work should not be conducted outside appropriately supervised facilities. Projects involving blood, body fluids, human tissue, unknown cultures, pathogens or other potentially hazardous biological materials require specific approval and may be prohibited.

14.4 Environmental responsibility

- Do not release chemicals, organisms or experimental materials into the environment without authorization.
- Minimize disturbance to habitats.
- Do not collect protected species or remove large quantities of natural material.
- Return field sites to a safe condition.
- Plan for disposal or reuse of project materials after the fair.

Approval comes first

your school has confirmed what consent and permission are required. Data gathered before approval cannot be used.

Leave the site as you found it

the field. Field work should end with the habitat undisturbed and the site in a safe condition.

15 / Intellectual Property and Disclosure

15.1 Why intellectual property matters

A science fair is a public learning environment. Students may create designs, software, devices or methods that they believe have future commercial or patent potential. Public presentation can affect intellectual property options in some countries. If a project may contain a valuable new invention, the student and school should discuss this before publicly sharing detailed technical information.

15.2 Do not disclose confidential information

Students should not include confidential information belonging to a company, university, school, family, mentor or other organization. Permission to use equipment or receive advice does not automatically give permission to publish private data or technical information.

15.3 Ownership

Ownership can depend on who created the work, school rules, agreements with sponsors and applicable law. This handbook does not determine legal ownership. Students and institutions should clarify ownership and permissions before accepting outside support that includes intellectual property conditions.

15.4 Acknowledging prior work

If your project improves an existing design, say so. Explain what existed before and what you changed. This strengthens rather than weakens the project because it shows that you understand the starting point.

Talk before you publish

Public display can affect patent options in some countries. If your design may be genuinely new, discuss it with your teacher and school before sharing drawings or code.

Access is not permission

Being allowed to use a laboratory, workshop or dataset does not give you the right to publish what you saw there. Ask the owner what you may include before you write it up.

PUT IT INTO PRACTICE

What I should discuss before sharing my work

Who can help me make that decision

16 / Project Management and Documentation

16.1 Keep a research or engineering notebook

Keep a dated research or engineering notebook on paper or in a school-approved digital format. It should show how the work developed over time.

16.2 What to record

- Date of each work session.
- Questions and ideas.
- Sources consulted.
- Sketches and calculations.
- Experimental setup.
- Raw data.
- Prototype versions.
- Problems and failed attempts.
- Changes to the method and reasons.
- Teacher or mentor advice.
- Photographs where permitted.
- Budget purchases and receipts where funding applies.
- Decisions made by the student team.
- Next actions.

16.4 Timeline management

Break the project into tasks and assign target dates. Include time for background research, approval, purchasing, pilot testing, full testing, analysis, report writing, poster preparation and practice. Build in some time for unexpected delays.

16.5 Team meetings

Teams should briefly record major decisions and responsibilities. If one person misses a task, the team should address it early rather than waiting until the science fair.

Your notebook is evidence

Keep a second copy

Back up data, photographs and code somewhere else — a school computer, a teacher's drive, printed data sheets. One lost phone or notebook should never end a six-month project.

17 / Preparing the Final Report

17.1 Purpose

Report what you actually did and learned. Explain differences from the proposal and confirm that changes stayed within approved safety and ethics requirements.

17.2 Recommended structure

Section	Purpose
Title page	Project title, students, school, mentor and category.
Abstract	A short summary of the problem, method, main result and conclusion.
Introduction	Context, problem and importance.
Background	Relevant prior knowledge and sources.
Question or design challenge	What the project set out to answer or achieve.
Objectives and hypothesis	As applicable.
Methodology	What was actually done, including changes from the proposal.
Results	Data, tables, figures and factual observations.
Analysis and discussion	What the results mean and how they compare with expectations or prior knowledge.
Conclusion	Direct answer to the question or evaluation against requirements.
Limitations	Important weaknesses or uncertainties.

Section	Purpose
Recommendations and future work	Reasonable next steps.
Impact and sustainability	Potential value and what may continue.
Acknowledgments	People or institutions that assisted.
AI disclosure	Material AI assistance, if any.
References	Sources cited.
Appendices	Detailed data, code excerpts, drawings or supporting documents when useful.

17.3 Figures and tables

- Give each figure or table a clear title.
- Label axes and units.
- Do not stretch graphs to exaggerate small differences.
- Explain the source of any external image or dataset.
- Use readable text.
- Do not include decorative graphics that make scientific information harder to see.

18 / Science Fair Display and Demonstration

18.1 Purpose of the display

Use a few key visuals and short explanations to show your question, process and results. The display should communicate the project quickly.

18.2 Recommended display sections

- Project title and student names.
- School and project category.
- Problem or research question.
- Background and why the issue matters.
- Objectives or design requirements.
- Method or design process.
- Key results with graphs, tables or photographs.
- Prototype or model where allowed.
- Conclusion.
- Limitations.
- Future work.
- Acknowledgments and selected references.

18.3 Display dimensions

Final board dimensions, table allocation, electrical access and venue restrictions will be stated in the official event instructions. Students must comply with the current event instructions even if their school uses a different local display format.

18.4 Items generally unsuitable for public display

- Open flames.
- Explosives or explosive components.
- Exposed hazardous chemicals.

- Unknown biological cultures.
- Human or animal body fluids.
- Unprotected sharp objects.
- Unsafe high voltage equipment.
- Pressurized systems without approved safeguards.
- Live animals unless specifically approved.
- Unlabeled substances.
- Personally identifying participant data.
- Food or drink intended for sampling by judges unless specifically approved.

18.5 Demonstrations

A demonstration should be short, reliable and safe. Students must be able to explain what the demonstration proves and what it does not prove. If a prototype cannot safely operate inside the venue, use a video of the approved test and present the physical prototype in a safe condition where permitted.

The few steps back test



19 / Oral Presentation and Judge Interview

19.1 The student should tell the story

Prepare a short explanation of your project, then be ready for questions. Judges want to understand your thinking.

19.2 A useful presentation structure

1. State the problem or question.
2. Explain why it matters.
3. Describe what you decided to do.
4. Highlight the method or design choices.
5. Show the most important results.
6. Explain what the results mean.
7. Discuss one limitation or challenge.
8. State what you would do next.

19.3 Questions judges may ask

- Why did you choose this problem?
- What did you know before you started?
- Which part of the project was most difficult?
- What variable did you control and why?
- How did you know your measurement was accurate enough?
- Why did you choose this design instead of another one?
- What was your most important result?
- What result surprised you?
- What would you change if you repeated the project?

- What work was done by your teacher or another mentor?
- Did you use AI, code libraries, online designs or external datasets?
- How much did the prototype cost?
- What evidence supports your conclusion?
- What are the limitations of your claim?

19.4 If you do not know an answer

Say that you do not know, then explain how you would find out. It is better to be truthful than to invent an answer. Judges often respect students who can identify the boundary of their knowledge.

19.5 Team interviews

All team members should participate naturally. Do not allow one person to answer every question. Judges may direct questions to different members to understand individual contributions and shared understanding.

Rehearse the conversation

Practise two things: a two minute explanation of your project, and answering questions you cannot predict. Ask a teacher, a classmate and someone outside science to question you. The outsider will find the sentences that only make sense inside your own head. Judges ask those.

20 / Judging Criteria and Scoring

20.1 Two stages of evaluation

The competition may evaluate projects at proposal stage and again at the final science fair. Proposal review determines whether a project is strong and appropriate enough to proceed or receive support. Final judging evaluates the actual completed work, student understanding and presentation.

20.2 Proposal review rubric from the Foundation source guidance

Criterion	What a strong proposal shows	Marks
Clear problem	One specific problem in a specific place, with evidence that it is real.	20
Workable plan	The plan fits the student's skills, time, school setting and available support.	25
Honest budget	Costs are itemized, realistic and mathematically correct.	20
Measurable results	The student states what will be different and how the result will be demonstrated.	20
Something that lasts	The project leaves useful knowledge, skills, evidence, equipment or a sustainable activity.	15
Total		100

20.3 Final science fair judging framework

Area	What judges consider
Problem and significance	Is the problem clear, meaningful and appropriately scoped?

Area	What judges consider
Scientific or technical understanding	Does the student understand the concepts behind the work?
Method or design quality	Is the approach logical, fair and technically appropriate?
Creativity and student contribution	What decisions, improvements or original thinking came from the student?
Evidence and analysis	Are results recorded, analyzed and interpreted appropriately?
Engineering performance where applicable	Was the prototype tested against meaningful requirements?
Limitations and honesty	Does the student recognize uncertainty, weaknesses and failed attempts?
Impact and feasibility	Is the claimed value realistic and supported?
Communication	Can the student explain the work clearly and answer questions?
Project documentation	Does the notebook, report and display support the authenticity and quality of the work?

21 / Awards and Recognition

21.1 Award structure

The exact award categories for each competition cycle will be announced by Jdiobe STEM Foundation. The Foundation may recognize overall excellence, category excellence and special achievements such as innovation, community impact, sustainability, technical communication or promising future potential.

21.2 Award Categories

1. Overall Science Fair Champion.
2. Category awards.
3. Most Innovative Solution.
4. Outstanding Engineering Design.
5. Outstanding Scientific Investigation.
6. Best Use of Data.
7. Environmental and Sustainability Impact.
8. Agriculture and Food Security Innovation.
9. Technology and Computing Innovation.
10. Outstanding Communication.
11. Promising Young Researcher.
12. Community Impact Recognition.

21.3 Awards do not replace learning

Awards recognize achievement. Feedback, new skills, confidence and the experience of completing a project are valuable outcomes for every participant.

What you keep afterwards

Awards are announced in one afternoon; the skills stay with you. Students leave able to define a problem, design a fair test, read their own data honestly and defend their reasoning to an expert. Ask for judge feedback whether or not your project is named.

PUT IT INTO PRACTICE

What I have learned so far

How I can build on that learning

22 / If Your Project Receives Foundation Funding

22.1 Funding agreement

Before funds are released, the student project and institution may be required to sign a short agreement describing the approved amount, purpose, reporting expectations, safeguarding conditions and financial responsibilities.

22.2 Financial records

- Keep receipts for approved purchases.
- Record the date, item, supplier and amount.
- Do not move money between major budget categories without required approval.
- Do not use project funds for personal expenses.
- Return unspent funds or request written approval for an alternative use when the funding agreement requires it.

22.3 Progress update

Funded projects may be required to submit a short midpoint update. The update should state what has been completed, what remains, what has changed, how much has been spent and whether the student needs guidance.

22.4 Final report

A final report is due within two weeks of project completion when required by the funding agreement. It should explain what was done, what was learned, what results were obtained, what changed from the original plan and what the student would do differently.

22.5 Tell us when a project changes

Projects sometimes encounter unavailable materials, failed experiments, changing school schedules or other difficulties. Reporting a genuine problem

is not misconduct. Hiding significant changes, using funds for unapproved purposes or inventing progress is a serious problem.

Every shilling, on paper

recording as you go, and missing receipts can delay further support. Report a changed price early.

23 / Common Reasons Projects Lose Marks

- The project is copied or the student cannot explain the work.
- The problem is described only in general terms with no evidence from the actual setting.
- The project is far too large for the available time, budget or student skill level.
- There is no meaningful measurement or test.
- The conclusion makes claims much larger than the evidence supports.
- The budget does not add up or contains unexplained costs.
- The student has no research notebook or raw data.
- A teacher or adult appears to have performed most of the technical work.
- The student hides failed trials or changes data to look better.
- The project uses unsafe procedures or begins restricted work without approval.
- Human participant privacy or consent is not handled appropriately.
- The project uses AI generated content or code that the student cannot explain.
- Sources are missing or copied material is not acknowledged.
- The display is attractive but the scientific or engineering content is weak.
- The prototype is shown but never meaningfully tested.
- The team cannot explain who contributed what.
- Sections of the proposal are left empty or answered with vague statements.
- Required teacher or school approval is missing.

Marks are lost quietly

Check your own project twice

Read this list with your mentor a month before submission, then again before the fair. Mark each line done, not applicable or needs work. Fix weaknesses while time remains.

If you cannot explain it, cut it

Any formula, line of code, graph or phrase you cannot explain in your own words will surface in the judge interview. Learn it properly, replace it, or take it out.

24 / Frequently Asked Questions

Do I need an expensive project?

No. Cost is not a measure of quality. A focused project with good evidence can be excellent using simple materials.

Can I use an idea I saw online?

You may learn from existing ideas, but simply copying a project is not acceptable. Acknowledge the source and develop your own question, comparison, improvement or application.

Can my teacher correct my English?

Your teacher may help you notice unclear grammar or organization, but the ideas and writing must remain your own.

Can a university or company mentor help me?

Yes, when the help is appropriate, disclosed and does not replace student ownership. The student must understand and explain all work presented.

What if my experiment does not work?

Record what happened. Diagnose possible causes, improve the method where safe and explain what you learned.

Can I change my project after approval?

Minor changes may be normal. Major changes involving the research question, budget, safety, ethics, human participants or hazardous procedures should be discussed with your teacher and may require Foundation approval.

Can I use AI?

Responsible assistance may be allowed, but AI must not replace your work or understanding. Disclose material AI use and verify outputs.

Can more than one team from a school participate?

The number of projects allowed per school will be stated in the current competition announcement.

Can I survey people?

Possibly, but you must obtain the required school and competition approval before collecting data, and you must protect privacy and consent.

Can I test a medicine or treatment on people?

No ordinary student project should administer unapproved treatments or perform invasive medical testing. Health related projects require strict ethical boundaries.

Can I bring live chemicals to the fair?

Only materials specifically permitted under official display rules. Hazardous substances are generally not suitable for public display.

Can my parent build the prototype?

A parent may provide safe practical assistance when a task is beyond what a student is permitted to do, but the student must design the project, make the key decisions and disclose significant help.

What happens if my proposal is not selected?

Where the competition process permits, feedback may help you improve. A rejected proposal is not a statement that you are not capable of doing STEM.

Who owns equipment bought with Foundation funds?

Ownership and custody will be governed by the relevant funding agreement and school arrangements. Do not assume equipment becomes personal property.

Can I publish my project online before the fair?

Usually this is a school and competition decision, but students with potentially patentable inventions should consider intellectual property implications before public disclosure.

PUT IT INTO PRACTICE

A question I still need to ask

Who can help me answer it

25 / Glossary

Abstract	A short summary of the complete project.
Accuracy	How close a measurement is to the true or accepted value, where that value is known.
Beneficiary	A person or group that gains value from the project.
Budget	An itemized plan for project spending.
Calibration	Checking or adjusting an instrument using a known reference or standard.
Constraint	A limit that an engineering design must respect, such as cost, size or available power.
Control	A comparison condition used to help interpret the effect of a tested factor.
Correlation	A relationship in which two measured variables change together. Correlation alone does not prove causation.
Data	Recorded observations, measurements or information used as evidence.
Dependent variable	The outcome measured in response to a change.
Design requirement	A measurable statement of what an engineering solution should achieve.
Engineering design	A structured process of defining a need, creating solutions, testing them and improving the design.
Evidence	Information used to support or challenge a claim.
Feasibility	Whether a project can realistically be completed with available time, resources, skills and approvals.
Hypothesis	A testable prediction based on reasoning or prior knowledge.

Independent variable	The factor deliberately changed in an experiment.
Indicator	A measure used to show whether an objective has been achieved.
Iteration	A repeated cycle of testing, learning and improving.
Limitation	A factor that restricts what a study or design can prove or achieve.
Methodology	The planned or actual process used to conduct a project.
Objective	A specific result the project intends to achieve.
Plagiarism	Using another person's words, ideas or work without proper acknowledgment.
Prototype	An early working version of a design used for testing and improvement.
Qualitative data	Descriptive information about qualities or characteristics.
Quantitative data	Numerical measurements or counts.
Research question	A focused question that the project will answer using evidence.
Risk	The possibility that a hazard may cause harm, considering likelihood and severity.
Sustainability	The ability of useful outcomes or practices to continue after initial project support ends.
Variable	A factor that can change or take different values.

JDIOBESTEM FOUNDATION

Contact Us for More Information

Phone: +256 706 097 533

Address:
P.O BOX 1234
Bugembe, Jinja, Uganda

www.jdiobestem.org